

Week 6 Book exercise 3.7)

Recall that we have a prior and likelihood as

$$p(\underline{w}) = \mathcal{N}(\underline{w} | \underline{m}_0, \underline{S}_0), \quad p(\underline{t} | \underline{X}, \underline{w}, \beta) = \prod_{n=1}^N \mathcal{N}(t_n | \underline{w}^T \phi(\underline{x}_n), \beta^{-2}).$$

We further note that the posterior will equal the prior times the likelihood up to a normalizing constant. In line with the idea behind the "completing the square" technique, we focus on the resulting exponential term

$$\begin{aligned} & -\frac{\beta}{2} \sum_{n=1}^N \{t_n - \underline{w}^T \phi(\underline{x}_n)\}^2 - \frac{1}{2} (\underline{w} - \underline{m}_0)^T \underline{S}_0^{-1} (\underline{w} - \underline{m}_0) \\ &= \frac{\beta}{2} \sum_{n=1}^N \{t_n^2 - 2t_n \underline{w}^T \phi(\underline{x}_n) + \underline{w}^T \phi(\underline{x}_n) \phi(\underline{x}_n)^T \underline{w}\} \\ & \quad - \frac{1}{2} (\underline{w} - \underline{m}_0)^T \underline{S}_0^{-1} (\underline{w} - \underline{m}_0) \end{aligned}$$

$$= -\frac{1}{2} \underline{w}^T \left[\sum_{n=1}^N \beta \phi(\underline{x}_n) \phi(\underline{x}_n)^T + \underline{S}_0^{-1} \right] \underline{w}$$

see (2.71) $\uparrow$

$$= -\frac{1}{2} \left[-2 \underline{m}_0^T \underline{S}_0^{-1} - \sum_{n=1}^N 2\beta t_n \phi(\underline{x}_n)^T \right] \underline{w} + \text{constant}$$

By comparison with the Gaussian distribution, it readily follows that

$$\underline{S}_N^{-1} = \underline{S}_0^{-1} + \beta \underline{\Phi}^T \underline{\Phi}, \quad \text{and}$$

$$\begin{aligned} -2 \underline{m}_N^T \underline{S}_N^{-1} &= -2 \underline{m}_0^T \underline{S}_0^{-1} - \sum_{n=1}^N 2\beta t_n \phi(\underline{x}_n)^T \\ \Rightarrow \underline{m}_N &= \underline{S}_N (\underline{S}_0^{-1} \underline{m}_0 + \beta \underline{\Phi}^T \underline{t}) \end{aligned}$$

Book exercise 3.10)

We know that

$$p(\underline{t} | \underline{w}, \beta) = \mathcal{N}(\underline{t} | \phi(\underline{x})^T \underline{w}, \beta^{-1}), \text{ and}$$

$$p(\underline{w} | \underline{t}, \alpha, \beta) = \mathcal{N}(\underline{w} | \underline{m}_w, \underline{s}_w).$$

Utilizing (2.113), (2.114), and (2.115), it holds that

$$p(\underline{t} | \underline{t}, \alpha, \beta) = \mathcal{N}(\phi(\underline{x})^T \underline{m}_w, \beta^{-1} + \phi(\underline{x})^T \underline{s}_w \phi(\underline{x})). \quad (*)$$

Noting that $\phi(\underline{x})^T \underline{m}_w = \underline{m}_w^T \phi(\underline{x})$,

(*) is equivalent to (3.58).

Linear regression (out-of-book)

Suggested numerical solutions are given on the following page.

1)

$$w = (\Phi^T \Phi)^{-1} \Phi^T t$$

$$\Phi(t) = \begin{bmatrix} 1 & t \end{bmatrix} = \begin{bmatrix} 1 & t_1 \\ 1 & t_2 \\ 1 & t_3 \end{bmatrix} \Rightarrow \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \left(\begin{bmatrix} 1 & 1 & 1 \\ t_1 & t_2 & t_3 \end{bmatrix} \begin{bmatrix} 1 & t_1 \\ 1 & t_2 \\ 1 & t_3 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 1 & 1 \\ t_1 & t_2 & t_3 \end{bmatrix} \begin{bmatrix} t_1 \\ t_2 \\ t_3 \end{bmatrix}$$

2)

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \left(\begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1.5 \\ 3 \end{bmatrix} = \left(\begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix} \right)^{-1} \begin{bmatrix} 5.5 \\ 6.5 \end{bmatrix} \Rightarrow \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 1.43 \\ 0.6 \end{bmatrix}$$

$$\left(\begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix} \right)^{-1} = \frac{1}{18-4} \begin{bmatrix} 6 & -2 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} 6/14 & -2/14 \\ -2/14 & 3/14 \end{bmatrix}$$

3,4)

$$\begin{aligned} x(t, w) &= w_0 + w_1 t \\ \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} &= \begin{bmatrix} 1.43 \\ 0.6 \end{bmatrix} \end{aligned} \Rightarrow x(t) = 1.43t + 0.6$$

